Aaradhya Dev Tamrakar

Embedded AI · Systems Engineering · Firmware & Edge ML

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Final-year Electronics & Information Engineering undergraduate building embedded ML software, firmware, and local systems utilities. ML and gateway lead for SPARK (ESP32-S3 wearable fall detection combining kinematic filtering with a quantized 18.5 KB INT8 CNN under FreeRTOS). Experienced in C/C++, TFLite Micro, microcontroller sensor interfacing, and native Windows systems utilities.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

RESEARCH & TECHNICAL PROJECTS

SPARK — Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TensorFlow Lite Micro · Python · FreeRTOS · MPU-6050

- Project lead for ML modeling, gateway processing, and explainability in a four-person team; hardware and firmware have dedicated owners.
- Trained 1D-CNN fall-detection models on the SisFall benchmark (38 subjects, 38,420 windows), evaluating subject-stratified cross-validation with **0.9185 AUC-ROC** and 87.81% sensitivity.
- Quantized model to an **18.5 KB INT8 CNN** footprint for ESP32-S3 edge deployment; on-device hardware testing, BLE streaming, and volunteer cohort validation remain pending.

Agent-Customization-Sync — Cross-IDE Configuration Tooling | 2025 – Present

Python · SHA-256 · CLI · Configuration Governance

- Built configuration snapshots, guarded restore, rules transpilation, and SHA-256 drift audits across development environments (Antigravity, Cursor, Claude Code).
- Audited instruction overhead and linked shared skills to reduce configuration duplication and maintain environment consistency across machines.

Gesture-Controlled Self-Balancing Robot (GCSBR) | 2024 – 2025

Arduino · C++ · MediaPipe Hands · CameraX · Kotlin · PID Control

- Implemented real-time PID stabilization firmware on an inverted pendulum robot, tuning balance control parameters for stable upright recovery.
- Developed an Android companion app (Kotlin, MediaPipe Hands, CameraX) mapping hand gestures to directional actuation commands over Bluetooth.

Win-Vault — Native Windows Folder Privacy Utility | 2025

Python · Win32 APIs · NTFS ACLs · PowerShell

- Engineered a zero-dependency Windows folder locking utility enforcing access restrictions via native NTFS Access Control Lists (ACLs).
- Eliminated third-party driver overhead by utilizing standard operating system security permissions and PowerShell automation.

NovaOptimizer / System-Optimizer — Workstation Memory Utility | 2025

Python · Win32 APIs · PowerShell · CLI

- Built a native Windows performance utility reclaiming **1.2–3.4 GB working-set RAM** under heavy multitasking developer workloads.
- Automated background cleanup routines via standard Windows process memory APIs without disrupting active IDE sessions.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Coordinate branch planning, technical workshops, and cross-committee communication for Kathmandu Engineering College IEEE student body.
- Facilitated technical awareness events and student engagement initiatives across engineering departments.

Makerspace Ambassador — KEC Maker's Space, Kathmandu Engineering College | June 2026 – Present

- Maintain website content, interface fixes, and documentation for student prototyping equipment.

- Support students with 3D printing, embedded electronics prototyping, and lab equipment access.

Event Manager (2026 – Present); Resource Manager (2024 – 2026) — Electronics Project Club (EPC), Kathmandu Engineering College | 2024 – Present

- Manage event logistics, hardware inventory, and project exhibitions for the college electronics department.
- Organized project showcases and hardware component distribution for student engineering competitions.

TECHNICAL SKILLS

Embedded Systems & Edge AI: C/C++, ESP32-S3, TensorFlow Lite Micro, FreeRTOS, Arduino, Sensor Interfacing (IMU, UART/I2C/SPI), Kotlin

Engineering Practice & Tooling: Git, GitHub, Linux, PowerShell, Pytest, Reproducible Configurations, WordCraft, LaTeX